Pennsylvania State Finances: 1820 to 1860*

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1 Summary

This document reports tables and figures that might find their way into a chapter on early state finance. New York might be the "good" case study. Pennsylvania might be the "bad" case study.

2 The Government Budget Constraints

The organizing principle is the government budget constraint with private assets (which we labeled "state funds" in the New York memo and call "state assets" in this memo).

$$\underbrace{G_t}_{\substack{\text{government}\\\text{spending}}} + \underbrace{r_t^B B_{t-1}}_{\substack{\text{interest on}\\\text{state debt}}} + \underbrace{A_t - A_{t-1}}_{\substack{\text{change in}\\\text{state funds}}} = \underbrace{T_t}_{\text{taxes}} + \underbrace{r_t^A A_{t-1}}_{\substack{\text{interest on}\\\text{state funds}}} + \underbrace{B_t - B_{t-1}}_{\substack{\text{change in}\\\text{state debt}}} \tag{1}$$

3 Pennsylvania Revenues and Expenditures

In figure 1, I plot total revenues and expenditures (including interest payments and defaulted payments). In figure 2, I plot gross and primary deficits

- Unlike New York, Pennsylvania included public works spending and revenues directly in its general account.
- The 1836-1837 spike in revenue is due to the federal government's payment of surplus federal funds to the states and unusually large fees received from bank charters.
- In figure 2 we see large deficits before the crisis. After the crisis, the Commonwealth ran a primary surplus and a balanced (gross) budget.

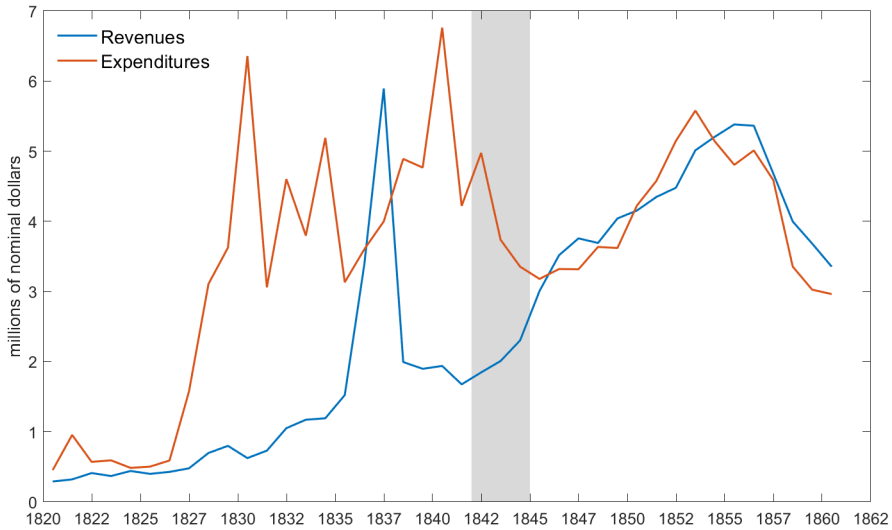

Figure 1: Pennsylvania State Revenue and Expenditures

The period of default (1842-1844) is shaded in grey.

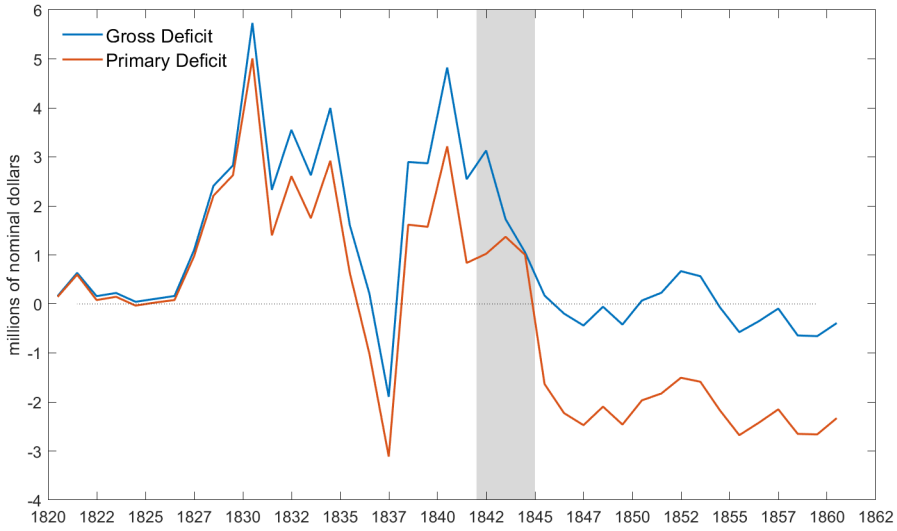

Figure 2: Pennsylvania Gross and Primary Deficits

The period of default (1842-1844) is shaded in grey.

In figure 3, I plot the sources of Commonwealth revenues from 1820 to 1860

- Land sales make up a relatively small share of total revenue. This is similar to New York State. Unlike New York, revenue from state assets is a small share of total revenue. See figure 12 of the New York State memo.
- The large revenues from the Other category in 1836 and 1837 are from "Enrollment of Bank Charters."
- The large payment from the federal government is the distribution of the federal surplus to the states mandated by the Deposit Act of 1836.
- Tax revenue increased starting in 1845, as a large number of new taxes were introduced.

In figure 4, I plot expenditure components over time.

- Spending on internal improvements and interest payments is far and away the largest spending categories.
- 1842, 1843, and 1844, Pennsylvania failed to make interest and other promised payments. As discussed below, bondholders were issued "interest certificates," and other non-payments to government contractors were recorded as "domestic creditors." These defaulted payments are reported in purple.
- Compare this figure to figures 8 and 11 in the New York State memo. In Pennsylvania, interest payments made up a much larger share of expenditures than they did in New York.

4 Debt and Assets

I plot the quantity of Pennsylvania State debt in figure 5, decomposed by loan purpose. I also split out the extraordinary "loans" issued during the crisis.

- Between the early 1825 and 1840, nearly all of the outstanding debt had been issued to fund internal improvements. Starting in the 1840s nearly all new debt was issued to rollover past debt or to make coupon payments. These refinancing loans allowed other revenue to be used to finance continued spending on internal improvements.

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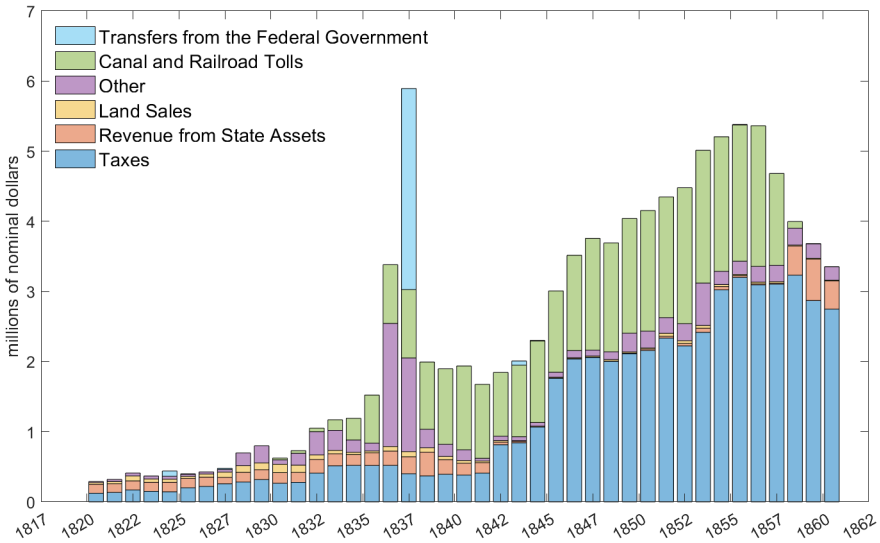

Figure 3: Pennsylvania Revenues

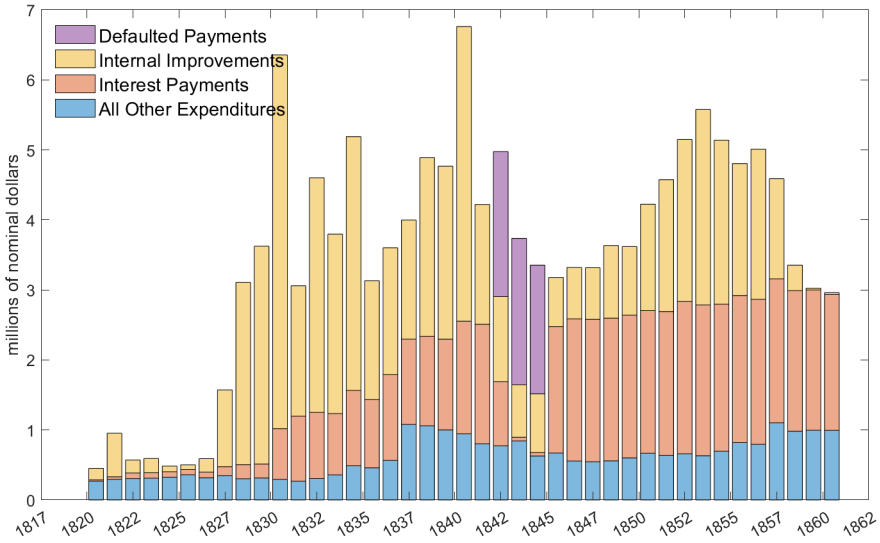

Figure 4: Pennsylvania Expenditures

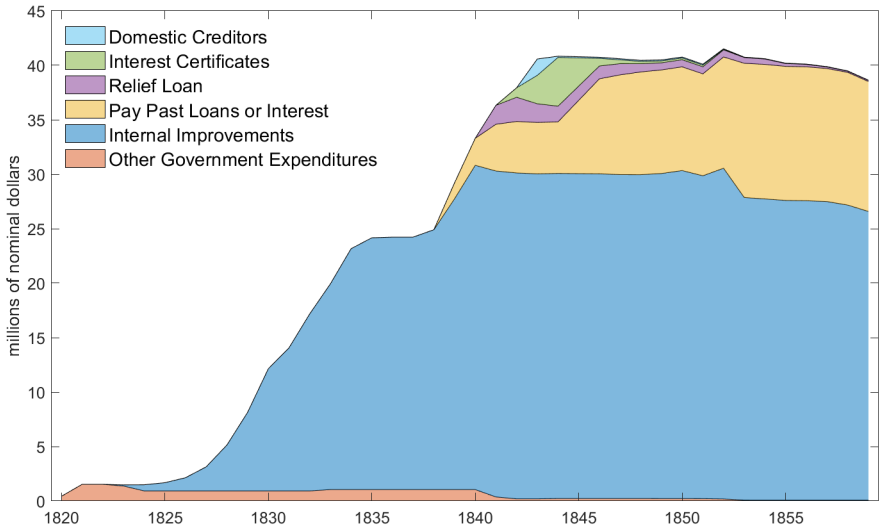

Figure 5: Pennsylvania Debt Decomposed by Purpose

- In 1840, Pennsylvania's debt stood at $33.3 million, and the State collected $1.9 million in total revenue. That same year, New York State's consolidated debt was $17.6 million, and it collected $2.9 million in total revenue (across its general and Canal accounts).

- From 1838 to 1842, the State Auditor General's and Treasurer's records of the State's debts included $2,867,514.78 as a debt to the federal government with the expectation that it would have to be repaid. Interest was never paid on this "debt," and the federal government never demanded its return. New York State never included this obligation in its debt records. We exclude this debt in figure 5.

- Pennsylvania owned bank stocks, turnpike and bridge stocks, and canal and navigation stocks. I plot the aggregate value of these stocks (reported at their par value) in figure 6. Dividends from these stocks are reported as "Revenue from State Assets" in figure 3.

- I assume, but have not been able to confirm, that the original source of funds for these stocks was Hamilton's 1790 assumption of the state debts.

- Starting in 1835, in the Annual Report of the State Treasurer, the State's canals, railroads, and bridges were recorded as State Assets and valued at cost. I have not included the value of these public works in figure 6.

- In 1843, the State sold $4,169,440.45 in face value of these stocks. They received $1,395,411.84 (1/3 of the face value) in this sale.

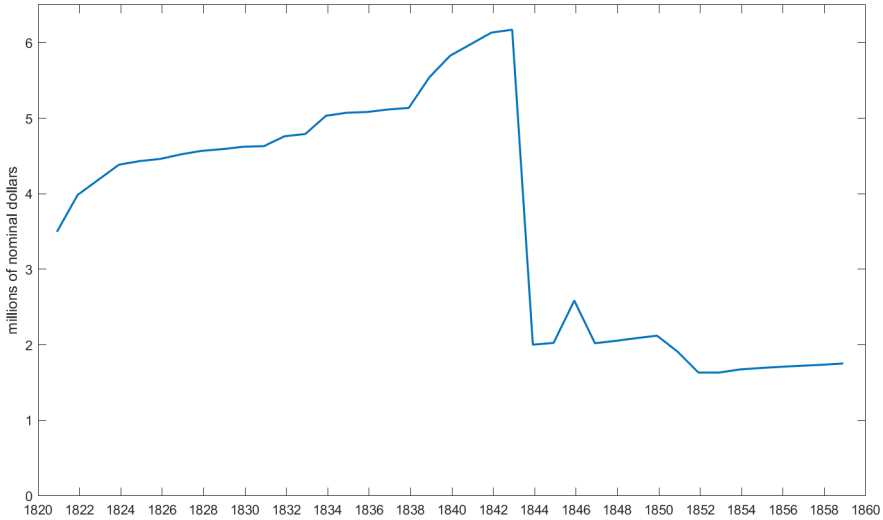

Figure 6: Pennsylvania State Assets, Not Including Public Works

5 Losses on the Canals and Railroads

- In response to the success of the Erie Canal, in February 1826, the Pennsylvania Legislature authorized the Main Line of Public Works to begin the building of canals from Philadelphia to Pittsburgh.
- Between 1830 and 1857, the State of Pennsylvania operated and collected toll revenue on nine divisions of the Pennsylvania Canal and two railroads.

	Length (in miles)	Cost of Construction
Railways		
Columbia and Philadelphia	82	$5,277,278
Allegheny Portage	36	2,708,672
Railway Total	118	$7,985,950
Pennsylvania Canal		
Eastern Division	45	$1,737,285
Juniata Division	128	3,575,966
Western Division	103	3,173,432
Delaware Division	60	1,454,937
Susquehanna Division	41	897,161
North Branch Division	73	1,598,379
West Branch Division	76	1,832,583
French Creek Division	49	817,780
Beaver Division	30	519,365
Canal Total	605	$15,606,888
Unfinished Improvements		$8,695,044
Other Costs		254,384
Grand Total		$32,542,268

Source: Pennsylvania Office of the Auditor General and Treasury (1854).

- The average cost to build the Pennsylvania Canal was $25,797 per mile.
- For comparison, the Erie Canal was 363 miles and cost $7,143,000 (or $19,678 per mile) to construct.
- In 1854, the State Treasurer and Auditor General detailed the expenditures on and revenues from these canals and railroads as well as the unfinished projects through 1853.

- For each of the 11 canals and railroad, they report annual disbursements "attendant consequent upon the operation of the works." These are "variable costs" for operating these canals and railroads and line up with numbers recorded in the annual reports of the State Treasurer, the Auditor General and the Canal Commission. These expenses include maintenance and wages paid to toll collectors, weigh-masters, and lock-keepers.

  This series of variable costs is reported in the red line (labeled Running Expenditures) in figure 7.

- They also report the annual revenue. This series lines up with the toll revenue reported elsewhere. I add canal fines to the toll revenue, and plot this series as the blue line in figure 7.

- The Treasurer and Auditor estimate that the total interest paid on interest for money borrowed to build and run these public works up to 1853 was $35.2 million. They don't provide an explanation of exactly how they compute these numbers. As we can see in figure 5, nearly all the Pennsylvania State debt was incurred to fund internal improvements.

  If we sum the coupon payments made on all the loans issued to pay for internal improvement or to pay principal or coupons on past loans from 1820 to 1853, we get $35.2 as well. Adding our estimate of these interest payments to the variable costs series produces the yellow line in figure 7.

- They also report the aggregate cost for building each of these canals and railroads. Summing these costs across the 11 public works comes to $32.5 million. If we sum all the spending on internal improvements reported elsewhere and net out the running expenditures, we get almost $30 million.[1] If we add our series of "All Other Spending" on internal improvements to the yellow line, we get the purple line in figure 7.

- These numbers are summarized in table 1.

- Even with our underestimate of the fixed costs of construction, it is clear that the losses on the Pennsylvania canals and railroads were large. The State spent either $87 million (their estimate) or $84 million (our estimate) to build, finance, and run these public works between 1820 and 1853. They collected only $25 million in revenue during this period.

- In figure 7, toll revenue (blue line) and running expenses (red line) track each other. After 1842, toll revenue consistently exceeded running expenses. Nevertheless, toll revenue comes nowhere close to covering the interest payments due – much less recouping the costs of construction.

[1] I don't know where other $2 million comes from.

	Official Numbers	Our Numbers
Expenses Incurred to Run the Canals and Railroads: 1830-1853	$19,447,654	$19,477,654
Interest Payments on Canal and Railroad Debt: 1827-1853	35,157,796	35,231,591
Cost to Build the Canals and Railroads: 1820-1853	32,542,268	28,954,372
Total Costs	$87,147,717	$83,663,617
Revenue from Tolls	$24,995,632	$24,995,632

Table 1: Aggregated Fixed Costs, Variable Cost and Revenue for the Pennsylvania Canals and Railroads

Source: Pennsylvania Office of the Auditor General and Treasury (1854).

- From 1842 to 1850, expenditures on internal improvements above those for running the canals and railroads were essentially zero.
- In 1844, the State offered a set of canals and railroads for sale at $20 million. There were no takers.
- Again, in 1854, the State put the railways and canals up for sale without success.
- Then, on June 25, 1857, the State sold the
  - Columbia and Philadelphia Railway
  - Allegheny Portage Railway
  - Eastern Division of Canal
  - Western Division of Canal
  - some other canals as well – still need to figure out the details

  to the Pennsylvania Railroad Company for $7.5 million.
- The following year, the remainder of the canals and railroads were sold to the Sunbury and Erie Railroad Company for $3.5 million. I don't see either of these payments in the accounts. Perhaps these payments show up in the 1860s.

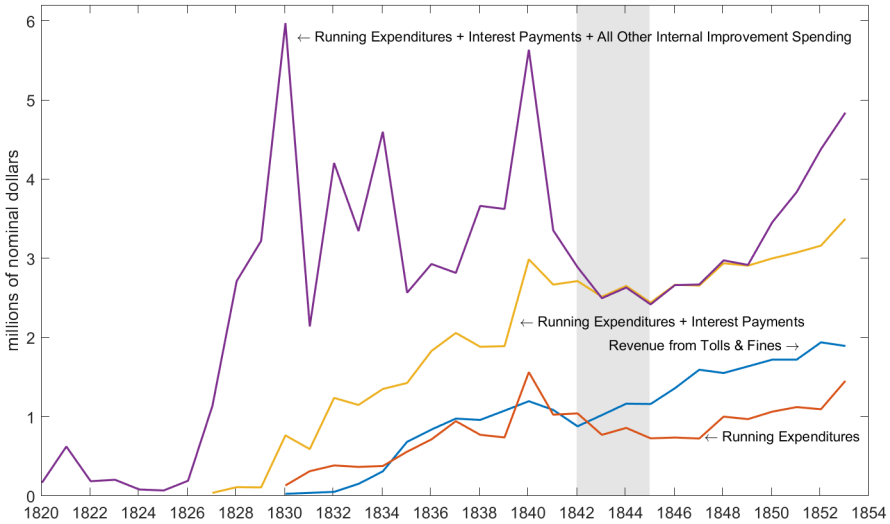

Figure 7: Canal and Railroad Revenue and Expenditures

The period of the default, 1842-1844, is shaded in grey.

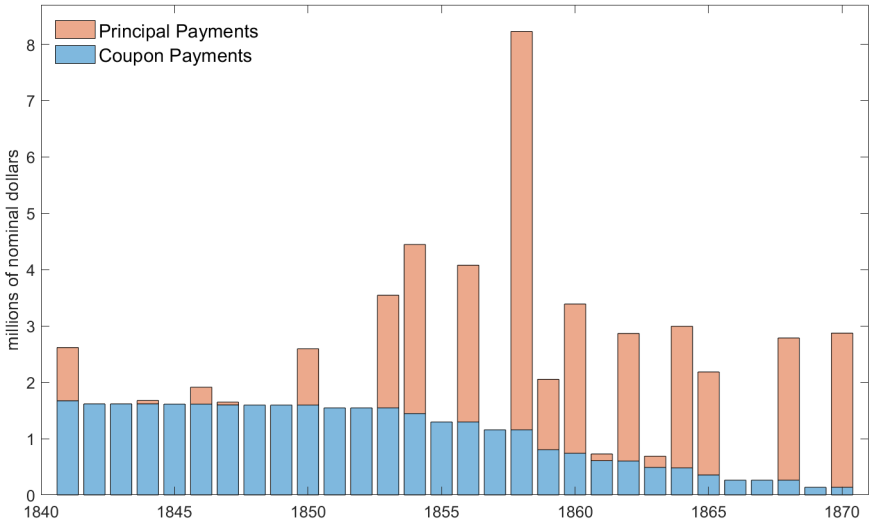

Figure 8: Debt Service Profile for December 1840

6 The 1841-1844 Crisis

In figure 8 I plot the debt service profile for the debt as of December 1840. Pennsylvania did not have a lot or principal payments due in 1840s. It was the promised coupon payments that killed'em.

6.1 1841: Relief Notes

- In 1838, 1839, and 1840, the State ran budget deficits of $2.9, $2.9, and $4.8 million.
- In 1841, the budget deficit was $2.5 million. Expenditures included
  - loan repayments $675,000
  - interest $1,707,000
  - internal improvements $1,710,000
  - other expenditures $803,000

  for a total of $4.2 million.

  That year, the State brought in only $1.7 million in revenue.
- They were unable to borrow from the bond market. I need to get the story.
- In 1841, Pennsylvania did not default on its promised interest payments or other obligations, but to do so, it had to borrow money from banks, which could then, in turn, issue small-denomination relief notes.
  - The Pennsylvania Relief Note Act of 1841 (formally the Act of May 4, 1841), an emergency measure passed by the Pennsylvania General Assembly, authorized a $2.22 million "relief loan." To secure the loan, the state authorized chartered banks to issue small-denomination paper currency known as "relief notes". These notes were issued in denominations of $1, $2, and $5, which were generally illegal under existing laws that banned small-denomination bank notes.
  - Banks that subscribed to the loan were permitted to issue these notes up to the amount of their subscription. In exchange, they were granted relief from certain state taxes and penalties for suspending specie payments.
  - The notes were made receivable for all debts due to the Commonwealth, effectively making them a state-backed currency.
  - The outstanding quantity of these notes is plotted in purple in figure 5.

6.2 1842-1844: Interest Certificates, Domestic Creditors, and Asset Sales

- The interest due in February 1842 was paid, but by August 1, the treasury was exhausted.
- English (1996) dates the default as August 1842. That lines up. He dates the resumption as February 1845.
- The Commonwealth was unable to borrow the $870,000 authorized by the Act of July 27, 1842. Need to explain the inability to borrow. Clearly they were shut out of the bond market in 1841.
- The State Treasurer issued to bondholders certificates of stock equal to the amount of interest then due. These certificates bore interest at five percent and were payable on August 1, 1843. See the green region in figure 5.
- The act also provided for the partial payment of claims held by contractors and others employed on the public works. These claimants were classified as *domestic creditors*. See the light blue region in figure 5.
- County commissioners were instructed, upon collecting county rates and levies, to impose an additional tax of one mill on the dollar on both real and personal property for the use of the Commonwealth. This provision was intended to remain in force for one year only.
- The governor was further directed to solicit proposals for the sale of the public works and to submit such proposals to the legislature. These efforts failed to generate any buyers.
- At the next interest date, February 1, 1843, the treasury again lacked sufficient resources to meet its obligations, and provision was once more made for the issuance of stock in payment of interest.
- In April 1843, a commission was appointed with authority to sell any or all state stock held by the Commonwealth in incorporated companies. All of the state's bank stock and other holdings were subsequently sold, The face value of these stocks was $4,169,440.45. The State received $1,395,411.84 (1/3 of the face value) in this sale. See figure 6.
- During the same year, the Commonwealth also received $60,313.27 from the federal government as its share of proceeds from the sale of public lands.
- I don't understand the resumption of payments. Relief notes, Interest certificates, and unpaid domestic creditors and a few ($800) domestic creditors are still outstanding in 1860. Were some creditors given priority over others?

7 Ownership of Pennsylvania State Debt

- Promises needed to be broken. The State could:
  - fail to pay domestic creditors,
  - fail to pay foreign bondholders, or
  - raise taxes on its citizens
- This led to a discrimination problem that was an echo of the one Hamilton faced in 1790:
  - Failing to pay domestic creditors or raising taxes risked immediate electoral backlash and more social unrest.
  - Failing to pay foreign bondholders carried reputational costs, but those costs were diffuse and delayed.
- I don't have the full story of this discrimination was resolved. There was no formal restructuring of the debt as far as I can tell.
- In 1842, there was $34,454,356.47 in Pennsylvania debt outstanding.

Pennsylvania	$9,635,613.47
Other US citizens	1,080537.60
Foreigners	23,738,206.00
Total	$34,454,356.47

Source: Pennsylvania Office of the Auditor General (1842).

The top five other states were

New York	$417,856
Massachusetts	129,000
District of Columbia	86,220
Virginia	84,700
Delaware	76,180

Source: Pennsylvania Office of the Auditor General (1842).

The top five foreign countries were

England	$20,026,458
Holland	1,822,266
France	570,000
West Indies	563,161
Switzerland	239,677

Source: Pennsylvania Office of the Auditor General (1842).

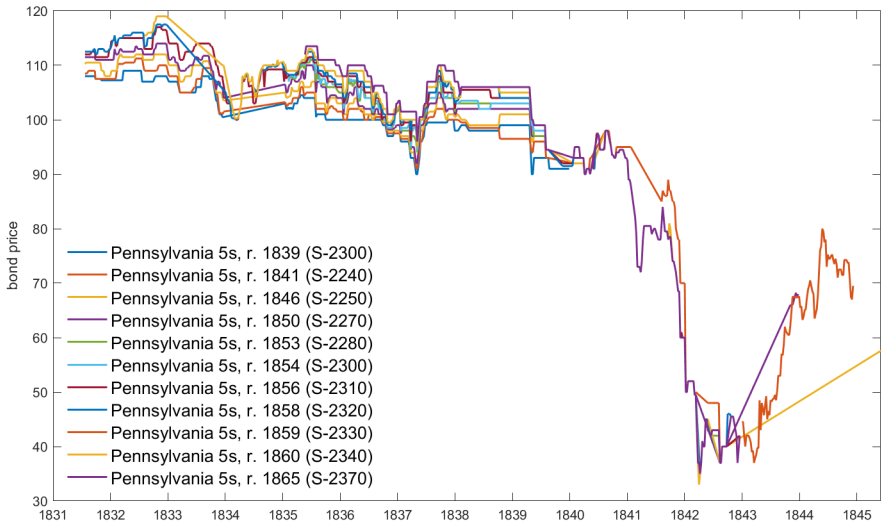

Figure 9: Prices of Various Pennsylvania State Bonds

Source: Sylla et al. (2006).

8 Bond Prices

9 Unit of Account Issues

- In May 1837, banks in Pennsylvania and across the United States suspended specie payments in response to the panic and financial crisis of that year.
- In July 1839, the legislature passed a resolution that the State would pay interest in "lawful money" (which I think in this case meant specie) and if the Treasury paid interest in any currency of less value than legal money, it must make up the difference. We see evidence of this "making up the difference" in the Treasury records.
- There is probably more to this story. Worthington (1887) has a discussion of this on page 47.

10 Data Sources

- The data for revenues, expenditures, state balances, and debts are from various issues of the *Report on the Finances of the Commonwealth of Pennsylvania for the Year 18XX Made to the Legislature by the Auditor General*.
- For some years, the same data are also reported in the *Report of the State Treasurer upon the Finances of Pennsylvania*.
- Unlike other state fiscal datasets, our records are consistent with the budget constraints presented at the start of this memo.

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